

1 (Hypergeometric distribution). Suppose we have an urn with w white balls and b black balls.

- (a) How many ways are there to draw n balls from this urn?
- (b) How many of these draws contain exactly k white balls?
- (c) Suppose we draw n balls out of this urn and X denotes the number of white balls. What is the p.m.f. of X ?

Solution. (a)

$$\binom{w+b}{n}$$

(b)

$$\binom{w}{k} \binom{b}{n-k}$$

(c) The p.m.f. of X is

$$P(X = k) = \frac{\binom{w}{k} \binom{b}{n-k}}{\binom{w+b}{n}}, \quad 0 \leq k \leq n$$

and such a random variable is said to be a *hypergeometric* random variable.

□

2 (Geometric distribution). Suppose that $X \sim \text{Geom}(p)$ and $Y \sim \text{Geom}(q)$ are independent.

- (a) Find $P(X > Y)$
- (b) If $p = q$, find $P(X \geq kY)$ for arbitrary $k \geq 1$.
- (c) If $p = q$, show that the distribution of $X \mid X + Y$ is uniform. That is,

$$P(X = k \mid X + Y = n) = \frac{1}{n-1}, \quad k = 1, \dots, n-1$$

Try to give a “story proof.”

Solution. Lets begin with (a). We approach this by using independence

$$P(X > Y) = \sum_{y \geq 1} P(X > y) P(Y = y)$$

Recalling the c.d.f. and p.m.f. of geometric random variables we conclude

$$\begin{aligned} P(X \geq kY) &= \sum_{y \geq 1} q(1-p)^y (1-q)^{y-1} \\ &= q(1-p) \sum_{y \geq 0} [(1-p)(1-q)]^y \\ &= \frac{q(1-p)}{1 - (1-p)(1-q)} \end{aligned}$$

which is what we wanted to find.

Onto (b), which is similar. As before we condition on the values of y and use independence

$$\begin{aligned} P(X > Y) &= \sum_{y \geq 1} P(X > ky) P(Y = y) \\ &= \sum_{y \geq 1} (1-p)^{ky-1} (1-p)^{y-1} p \\ &= p(1-p)^{k-1} \sum_{y \geq 1} (1-p)^{y(k+1)} \\ &= \frac{p(1-p)^{k-1}}{1 - (1-p)^{k+1}} \end{aligned}$$

Lets do (c). Note that by independence

$$P(X = k, Y = n - k) = P(X = k)P(Y = n - k) = p^2(1-p)^{n-2}$$

does not depend on k , and since $X + Y = k$ implies that

$$(X, Y) \in \{(k, n - k) : 1 \leq k \leq n - 1\}.$$

Since $X = k$ narrows it down to one of these equiprobable events, we conclude that

$$P(X = k | X + Y = n) = \frac{1}{n - 1}$$

as we wanted. □

3. Let n be a positive integer and

$$F(x) = \begin{cases} 0 & x < 0 \\ \lfloor x \rfloor / n & 0 \leq x \leq n \\ 1 & x > n \end{cases}.$$

Show that F is a c.d.f and find the p.m.f. that it corresponds to.

Solution. Clearly F is increasing, $F(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $F(x) \rightarrow 1$ as $x \rightarrow \infty$. Moreover, F is continuous everywhere outside of $\{0, \dots, n\}$ and is right-continuous inside $\{0, \dots, n\}$

$$\lim_{x \rightarrow k^-} F(x) = k/n = F(k).$$

This confirms that its a valid c.d.f.

The associated p.m.f. can be obtained by

$$p(k) = F(k) - F(k - 1) = \frac{k}{n} - \frac{k - 1}{n} = 1/n$$

which means that this is the c.d.f of a uniform random variable. □